

ISAAC AKOWANOU

Computer Science Student | Lubbock, TX | akowanouisaac@gmail.com | [GitHub](#) | [LinkedIn](#)

EDUCATION

Texas Tech University

Bachelor of Science in Computer Science

GPA: 3.89

Lubbock, TX

Expected Spring 2029

Collin College

Associate of Science, General Studies

McKinney, TX

May 2026

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, SQL, HTML/CSS

Frameworks/Tools: FastAPI, React, SQLAlchemy, PostgreSQL, Alembic, TensorFlow/Keras, Pandas, Git/GitHub

PROJECTS

School Management System | FastAPI, React, SQLAlchemy | [GitHub](#)

- Built a bilingual school-management platform with 133 REST API operations and 39 React pages for role-specific administrator, teacher, and parent workflows.
- Designed a 25-model relational schema with 32 Alembic migrations supporting multi-year records, report snapshots, audit history, and recoverable soft deletion.
- Implemented a deterministic Beninese /20 grading engine across 3 trimesters with coefficient-weighted results and French, English, and bilingual averages.
- Developed 628 backend tests across 50 files covering authentication, authorization, grading, reporting, migrations, imports, and database integrity.

MiniShell - Unix Shell Implementation | C | [GitHub](#)

- Built a ~540-line Unix shell in C with built-in commands, I/O redirection, pipelines, background execution, and a 100-command history.
- Implemented process execution with `fork()`, `execvp()`, and `wait()`; used `pipe()` and `dup2()` for IPC and file-descriptor routing, separating built-in command handling from forked processes.

Pneumonia Detection CNN | Python, TensorFlow | [GitHub](#)

- Designed a custom 4-block CNN ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ filters) with batch normalization and dropout (~7M parameters), achieving ~90% accuracy, ~89% precision, ~92% recall, and ~91% F1-score on 5,863 chest X-rays.
- Used augmentation and regularization for class imbalance and overfitting; implemented Grad-CAM visualizations for interpretability.

Job Market Analyzer | Python | [GitHub](#)

- Built a Python/Pandas pipeline for 30 tech-job records — standardizing data, removing duplicates, parsing salary ranges, and flagging remote roles — then exported cleaned CSV results and a 6-panel dashboard.

EXPERIENCE

Collin College - Facilities & Construction Department

Student Intern (19.5 hrs/week)

McKinney, TX

June 2025 - July 2026

- Supported department operations through data entry, digital documentation, record management, and spreadsheet-based tracking.
- Organized and verified large volumes of electronic records while maintaining consistent filing systems and confidentiality.

HONORS & ENGAGEMENT

President's List, Collin College - Fall 2024, Spring 2025 & Fall 2025

Phi Theta Kappa (PTK) — Member, Fall 2024–Spring 2026